

BEING COMFORTABLE WITH OUR PERFORMANCE



Fredrik Hamberger,
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Intel talks about their dominant market position in the mobile processor space for gaming notebooks

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Q OEMs produce a limited number of laptops in a launch cycle. Now, AMD and Apple are exploring using ARM processors in their laptops. Considering these factors how does Intel plan on adapting to this new market dynamic?

Fredrik: I think if you look at things such as our performance leadership and our focus on optimising around gameplay then you can see why we're so confident in our lineup, it's number one. Also, our engagement with the ecosystem, both eSports as well as with component vendors and PC manufacturers is unlike others. So, we have an ecosystem, tuning and deep engagement. You can also look at the scale that we bring with the volume designs. All these factors cover a lot of the choices that customers want in the market. Then you get to our platform and you look at the performance which are the fastest frequencies on anything out there today. Even our Core i5 processors have higher frequencies than anything else except for the rest of our stack. So, we're very confident in our position. And we expect to be aggressive in how we go to market with it. We're also going to continue to use our 9th Gen for more entry-level gaming configurations. Those designs already exist so we should see the 9th Gen continue in the market to address those low price

points and then the 10th Gen will sit on top of that. So, our portfolio looks very healthy. Competitively, of course, once we start getting benchmarks from out there, and reviewers like yourself can get their hands on the products, you'll see why we're very comfortable with our performance, especially gaming performance.

Q Having a higher frequency leads to reduced battery life. The Comet Lake-H processors seem to have really high frequencies, so, could you elaborate on how long these new processors can retain their peak performance state and what kind of battery-life compromises does this higher frequency lead to?

Fredrik: The CPU is a small ingredient of the total system's power consumption. So, if you're playing games, or you're doing high-end 4k editing, or video rendering, or exports; you're most likely plugged into the wall. Roughly 80 per cent of the time they're plugged in. Because people want to get that best level of performance. So that's number one. Number two, we do a tonne of work on ensuring that when not doing high performance tasks, the system is very efficient. If you're looking for the richest possible experience, and you want to do that on battery, we'll do the best to sort of deliver it to you. So, it's really dependent on you to take advantage of the power when you want it. But things that we worked on a lot with the ecosystem is our adaptive tuning and system optimisation together with OEMs. There's also panel types of specs that we've been working on such as variable refresh rates. So, if you're not running a game

at 120Hz, or there's no reason why it should run at 120Hz, it scales down to lower power states. There's modern standby and all those types of tricks and things that we've been working on for a long time. We have our resources engaged in helping OEMs implement that in their systems. You hear a lot about that with Project Athena, where we are helping the ecosystem move to more efficient solutions. But the main theme is really around partnering with OEMs to build the most efficient systems, tune them together and implement a lot of these tools to make sure that, when the system is not running at that highest performance mode, it's very efficient. Our data says that around 70 to 80 per cent of the time, on an H series notebook people are doing the same everyday tasks such as watching movies, social productivity, etc. That's where we think we can get the biggest bang for the buck. Let's take another example now, if you are running an HDR 1000 notebook in the office at high or maximum brightness when you're on battery, then it's going to consume it rapidly. So it's really dependent on the battery. It also largely depends on the use case that a customer wants. So, if you're looking for that high-end, fastest performance use case, it will obviously consume power. But the goal is when you're looking for battery efficiency, we give you that best possible experience. It is all about giving the end user choices, each of which are good choices. 